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PTSD's true color: Examining the effect of coloring on anxiety, stress, and working memory in veterans

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ABSTRACT

Given the high prevalence of PTSD in veterans (up to 30%), it is important to consider alternative treatments targeted at reducing some of the negative mental health symptoms, such as anxiety and stress. Coloring, and art therapy more generally, has been reported to be one of the most effective treatments in reducing the symptoms of posttraumatic stress disorder. Previous research on the benefits of coloring have focused on samples comprised of undergraduate students or children, rather than a population who suffer from anxiety disorders. Thus, the aim of the present study was to extend this existing research and investigate effects of coloring activities in veterans with PTSD. The present study included mandalas for structured coloring compared with free-drawing as an unstructured coloring activity. The findings revealed several key patterns: (1) veterans with PTSD showed decreased anxiety in the structured coloring condition, but not free-form coloring condition; (2) veterans with PTSD showed decreased stress in both structured and free-form coloring conditions; (3) both those with and without PTSD had higher working memory scores in the free-form coloring condition. While the coloring and drawing activities in this study do not constitute art therapy, the results suggest that these simple activities that can be done on an individual basis may yield both mental health and cognitive benefits, especially for veterans with PTSD.

1. Introduction

Post-Traumatic Stress Disorder (PTSD) is a debilitating anxiety disorder which can occur after someone has experienced a traumatic event in their lives. PTSD stressors are characterized by the belief that someone's life is in danger, as well as the belief that the person has no control over what is occurring (Bisson et al., 2007). The probability of developing PTSD depends on factors such as the intensity of trauma, personal strength of reaction to the event, and the amount of control felt during the event. PTSD does not affect everyone who has experienced a traumatic event, but it can develop after single exposure to trauma. This disorder can become severe enough that it prevents normal functioning.

The national prevalence rate is approximately 7% to 8%, with certain populations presenting a greater risk due to genes, exposure, or stressor type ("Post-Traumatic Stress Disorder,", 2017; Prins et al., 2003). For example, veterans have PTSD rates ranging from two to almost four times higher than the general population, depending upon the war in which they served. A notable comparison between prevalence rates comes from veterans who were active during the Vietnam War compared to those who were active in the Gulf War: the lifetime prevalence rates of PTSD for men who participated in the Vietnam War) is 30.9% versus 12.1% for participants of the Gulf War (Gradus, 2016).

Given the high prevalence of PTSD in veterans, it is important to consider alternative treatments targeted at reducing some of the negative mental health symptoms, such as anxiety and stress. The present study explored the use of art therapy, specifically coloring, as a potential way to induce a state of mindfulness. Art therapy refers to facilitating the client's creativity in a therapeutic context and can involve both traditional (e.g., paints, crayons) and non-traditional (e.g., fabric, nature objects) art media (Fausek, 1997). Art therapy has been reported to be one of the most effective treatments in reducing the symptoms of posttraumatic stress disorder because it provides an avenue for the individual to address their fears in a nonverbal context (Appleton, 2001; Levin, Lazrove, & Van Der Kolk, 1999; Orr, 1997; Wollman, 1993). Studies have reported that creating a mandala representing specific emotions related to their trauma decreased posttraumatic stress symptoms in an experimental group versus a control group (Henderson, Rosen, & Mascaro, 2007). Free drawing was also more effective than a writing task in treating nightmares for veterans with PTSD (Morgan & Johnson, 1995).

Researchers have explored the efficacy of different types of coloring

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activities in reducing anxiety in non-clinical populations. Curry and Kasser, (2005), found that structured coloring (a circular mandala and an irregular plaid design) was more effective in reducing state anxiety, compared to a free-form coloring task (coloring a blank piece of paper). According to the authors, the benefits of structured coloring stems from two aspects. First, the structured coloring designs were sufficiently complex to maintain attention and interest, yet did not require excessive thought or focus. Second, both the mandala and plaid coloring designs provided a structure that may have helped organize the "inner chaos" that often accompanies anxiety. Van Der Vennet and Serice (2012) replicated this study and found that coloring a mandala was more effective than coloring a plaid design in reducing anxiety in their sample. Coloring mandalas were also found to be more effective in reducing state anxiety compared to free-form coloring, in school-aged children (Carsley, Heath, & Fajnerova, 2015).

While these studies provide insight into the benefits of coloring for anxiety reduction, their samples comprised of undergraduate students or children, rather than a population who suffer from anxiety disorders. Thus, the aim of the present study was to extend this existing research and investigate effects of coloring activities in veterans with and without PTSD. Given the high proportion of veterans that experience PTSD (up to 30%), an anxiety disorder, and in light of the efficacy of art therapy in reducing anxiety symptoms, it is an important issue to investigate. The use of mandalas for structured coloring in this present study was based on the previous findings of the efficacy of mandalas in reducing anxiety in non-clinical populations. Free drawing as a unstructured coloring activity was used as a comparison, also in line with previous studies (Carsley et al., 2015; Curry & Kasser, 2005; Van Der Vennet & Serice, 2012)

In addition to anxiety and stress, veterans with PTSD also demonstrate cognitive deficits, such as in working memory, the ability to process and remember information (Schweizer & Dalgleish, 2011). Some researchers suggest that working memory plays a critical role in goal-directed behavior by keeping current goals in mind, while filtering distractions and focusing attention (Soto, Heinke, Humphreys, & Blanco, 2005). There is a growing body of evidence that suggests patients with PTSD demonstrate limited working memory resources that affect cognitive functioning (e.g., Honzel, Justus, and Swick, 2014). A few studies have also examined the neural bases of working memory deficits in PTSD. Brain scans indicate that veterans with PTSD are less likely to have activation in their dorsolateral prefrontal cortex (DLPFC) when completing visual and verbal tasks that require monitoring and updating information, which is associated with working memory function (Ehlers et al., 2003; also, Weber et al., 2005). To date, there is no research on the potential benefits of coloring and working memory; however, art therapy in the form of doodling has been found to aid memory recall by reducing daydreaming (Andrade, 2009). Thus, a related aim of the present study is to explore whether structured or free-form coloring might also improve working memory.

2. Method

2.1. Participants

Twenty-four veterans (15 males; Range 21–49 years; M age 32.25 years, SD = 8.48 years) from a public Florida university took part in this study in exchange for extra credit or a $20 gift card. Participants were recruited through the University Military Veterans Resource Center and Disability Resource Center. The majority of participants were Caucasians ($n = 17$), but there also were Hispanics ($n = 3$), Native Americans ($n = 2$), and an Asian/African American participant ($n = 1$). All participants were classified as individuals who have served in the armed forces; 50% served for less than or equal to 5 years ($n = 12$), 25% of the participants served between 5 and 10 years ($n = 6$), and the remaining participants served between 11 and 20 years.

3. Materials

Primary Care PTSD Screen. The Primary Care PTSD Screen is a four-item screen that was designed for use in primary care and other medical settings, as well as a screener for PTSD at the VA. The screener includes an introductory sentence to cue respondents to traumatic events. A positive response to three or more items indicates that the test taker has symptoms of PTSD. This scale has good internal reliability with a Cronbach's alpha of 0.78 (Freedy et al., 2010; Prins et al., 2003).

Brief State-Trait Anxiety Inventory. The six-item State-Trait Anxiety Inventory-State was administered to assess state anxiety levels. All questions are measured on a four-point Likert Scale (1 = almost never; 4 = almost always). There is good internal consistency with Cronbach's alphas between 0.83 and 0.86, and it was also found to be highly correlated with the full 20-item State-Trait Anxiety Inventory (Tluczek, Henriques, & Brown, 2009).

Perceived Stress Scale. The Perceived Stress Scale was administered to measure the degree to which individuals appraise their life situations as stressful (Lee, 2012a). This scale consists of 10 questions in two areas: psychological competency and psychological vulnerability. Each question is rated on a five-point scale (0 = never; 4 = very often). Scores in each area are then summed to obtain an overall score in each of the two areas, psychological competency scores range from zero to twenty and psychological vulnerability scores range from zero to thirty-five. Higher scores indicate a greater degree of the measured construct, psychological competency or psychological vulnerability. Cronbach's alpha coefficients for these subscales were 0.80 and 0.85, respectively, and the correlation between these two factors was not statistically significant indicating good discriminant validity (Lee, 2012b).

Working Memory. Working Memory was measured using a modified version of the Backward Digit Recall from a standardized assessment, the Automated Working Memory Assessment (Alloway, 2007). The individual recalls a sequence of spoken digits in the reverse order. The test begins with recalling two numbers in backward order and increases by one item in each block, up to nine numbers per block. There were two trials in each block and the number stimuli were randomized for the different testing phases. Scoring was calculated based on the highest block (span) where they correctly recalled one of the two trials. Test reliability is 0.86 (Alloway, 2007).

4. Procedure

Each participant completed two days of testing on an individual basis, approximately one week apart. In Phase 1, participants completed the Primary Care PTSD screening, the Perceived Stress Scale, the Brief State-Trait Anxiety Inventory-S, and the Working Memory test. Phase 2 was the structured/free-form coloring conditions where participants were provided with 12 colored pencils and had 20 min to complete the task (see Curry and Kasser, 2005; Van Der Vennet and Serice, 2012). In the structured coloring condition, participants were provided with an outline of a mandala on 8.5″ x 11″ paper. In the free-form coloring condition, participants were given a blank piece of 8.5″ x 11″ paper. In both conditions, participants were instructed to color the paper in front of them for 20 min, using the colored pencils provided for them. In Phase 3, participants completed the Perceived Stress Scale, the Brief State-Trait Anxiety Inventory-S, and the Working Memory test again. On Day 2, the procedure was the same, with coloring conditions counterbalanced. Participants were debriefed at the end of Phase 3.

5. Data Analysis

Participants were divided into those who answered positively to three or more questions to the Primary Care PTSD Screen (PTSD group $n = 8$) and those who did not (non-PTSD group $n = 16$; Table 1). To investigate the impact of the different coloring/free draw conditions on stress, anxiety, and working memory, the data was analyzed using three

Table 1
Demographic characteristics of participants.

Demographic Characteristics	
% Male	63%
Age range (Years)	21 – 49
Median age (Years)	32.25
Age SD (Months)	102
% Caucasian	71%
% Hispanic	13%
% Native American	8%
% Asian / African American	4%
% Filipino / Greek	4%
% Who served 5 or less years	50%
% Who served 5 – 10 years	25%
% Who served 11 – 20 years	25%

Wilcoxon signed-ranked tests, with the data being split first by PTSD status, then Coloring Condition, and finally by both PTSD group and structured/free-form coloring conditions. This data was analyzed using SPSS, and data for two participants in the free draw coloring condition was missing due to attrition. This was believed to have a negligible effect on the data obtained.

6. Results

6.1. PTSD groups

The findings indicated that the PTSD participants showed a significant decrease in stress ($p < 0.001$, $Z = -3.69$), anxiety ($p = 0.011$, $Z = -2.53$), as well as a significant increase in working memory ($p = 0.001$, $Z = -3.19$). In contrast, the non-PTSD participants showed a significant decrease in anxiety ($p = 0.046$, $Z = -2.00$), but not in stress ($p = 0.341$, $Z = -0.95$) or working memory ($p = 0.190$, $Z = -1.31$).

6.2. Structured vs. free-form coloring conditions

The findings indicated that structured coloring condition led to a significant decrease in stress ($p = 0.023$, $Z = -2.28$) and anxiety ($p = 0.008$, $Z = -2.65$), but no significant change in working memory ($p = 0.198$, $Z = -1.30$). Participant stress scores in the free-form coloring condition were significantly lower ($p = 0.026$, $Z = -2.23$) and were significantly higher for the working memory ($p = 0.001$, $Z = -3.34$), but showing no significant change in anxiety ($p = 0.198$, $Z = -1.29$).

6.3. PTSD groups x Coloring conditions

See Table 2. For individuals with PTSD, perceived stress scores were significantly lower in both the structured and free-form coloring conditions ($p = 0.005$, $Z = -2.84$ and $p = -0.019$, $Z = -2.35$, respectively). For individuals without PTSD, stress was not significantly lowered in either the structured or free-form coloring conditions ($p = 0.733$, $Z = -0.34$ and $p = 0.343$, $Z = -0.95$, respectively).

With respect to anxiety scores, participants with PTSD showed lower anxiety scores in the structured coloring condition ($p = 0.031$, $Z = -2.16$), but not in the free-form coloring condition ($p = 0.135$, $Z = -1.50$). There was not a significant change in anxiety scores for those without PTSD in either the structured or free-form coloring conditions ($p = 0.062$, $Z = -1.87$ and $p = 0.344$, $Z = -0.95$, respectively).

Finally, working memory scores were higher in the free-form coloring condition for those with PTSD ($p = 0.009$, $Z = -2.62$), but not in the structured coloring ($p = 0.076$, $Z = -1.77$). The pattern was the same for those without PTSD: the free-form coloring condition led to higher working memory scores ($p = 0.039$, $Z = -2.06$), but the structured coloring condition did not ($p = 0.932$, $Z = -0.09$).

Table 2
Descriptive statistics of pre-and posttest stress, anxiety, and working memory scores for PTSD and non-PTSD Participants.

PTSD Structured coloring	Mean	SD	Non-PTSD Mean	SD
T1 stress	18.25	3.85	15.31	6.63
T1 anxiety	12.50	2.20	10.44	3.33
T1 working memory	8.13	2.42	7.56	2.58
T2 stress	17.38	6.68	13.88	5.85
T2 anxiety	10.38	2.77	9.38	3.48
T2 working memory	8.13	3.00	8.44	2.31
PTSD Free form coloring	**Mean**	**SD**	**Non-PTSD Mean**	**SD**
T1 stress	17.14	4.02	14.07	6.39
T1 anxiety	11.71	3.35	9.93	4.40
T1 working memory	7.57	0.79	7.80	2.11
T2 stress	15.57	6.48	12.93	6.34
T2 anxiety	10.29	4.15	8.93	3.01
T2 working memory	9.00	1.63	9.07	2.58

7. Discussion

The present study revealed several key patterns: (1) veterans with PTSD showed decreased anxiety in the structured coloring condition, but not free-form coloring condition; (2) veterans with PTSD showed decreased stress in both structured and free-form coloring conditions; (3) both those with and without PTSD had higher working memory scores in the free-form coloring condition. These results will be discussed in turn.

We first look at the finding that structured coloring reduced anxiety in veterans with PTSD. Small studies on art therapy, which is often used as a catalyst to create a coherent narrative to understand and process traumatic memories (Van der Kolk & Fisler, 1995), have found this visual form of therapy to yield the most benefits for veterans with severe PTSD symptoms compared to other forms of therapy (e.g., group therapy, drama, journal; Johnson, Rosenheck, and Fontana, 1997). While structured coloring in the present study was not combined with direct therapeutic intervention, it may have involved some therapeutic mechanisms. Art making in general is thought to reduce arousal by encouraging a relaxing and meditative state, regardless of whether the art is related to the traumatic experiences (Collie, Backos, Malchiodi, & Spiegel, 2006). In the present study, it is possible that the use of a mandala for the structured coloring condition was therapeutic because it provided both a pleasurable distraction from negative thoughts and emotions (see Johnson et al., 1997). The circular structure of the mandala may also have been indirectly beneficial as it is often associated with a meditative tradition to encourage a sense of wholeness, as well as provided a way to organize the inner chaos that is symptomatic of anxiety (Curry & Kasser, 2005).

The finding that structured coloring can decrease anxiety advances current research in two ways. First, it extends previous research on art therapy to suggest that a simple structured activity such as coloring a mandala for 20 min can provide some mental health benefits for veterans with PTSD. Secondly, we found that structured coloring can reduce anxiety in veterans with PTSD, not just college students (Curry & Kasser, 2005; Van Der Vennet & Serice, 2012). In contrast, free-form coloring did not reduce anxiety, which is in line with Curry and Kasser's findings with college students. They suggested that the unstructured nature of drawing may in fact have induced anxiety due to the lack of direction. Indeed, our observations of the veterans during the free-form coloring condition suggested that the veterans reported increased anxiety from not knowing what to draw. For example, one veteran asked to stop coloring early, stating that he had finished with his picture, while another veteran wrote out his weekly schedule instead.

The present study also found that stress decreased in both structured and free-form coloring conditions for veterans with PTSD. The production of art is thought to be stress-reducing (Bell & Robbins, 2007) and art therapy in general is useful in teaching stress management to veterans (see Collie et al., 2006). However, despite a growing number of “stress-relief” coloring books and anecdotal claims, no studies have directly investigated the potential stress-relieving benefits of coloring in clinical populations. The present study found that both structured and free-form coloring had stress-relieving benefits. This finding is in line with studies using trauma focused art therapy in PTSD populations. Chapma, Morabit, Ladakako, Schreier, and Knudson (2001) reported that children receiving the art therapy intervention showed a reduction in acute stress symptoms. Lyshak-Stelzer, Singer, St John, and Chemtob (2007) also found that trauma focused art therapy led to greater reduction in PTSD symptom severity compared to standard arts-and-craft-making activity. The authors suggest that using art products to share traumatic experiences utilizes displacement in a constructive way. In the present study, although the art forms did not include a trauma focused element, it is possible that coloring, in both structured and unstructured forms, provide the individual the opportunity to orient to safe visual representations, thereby decreasing their perceived stress levels.

One explanation for the reduction in state-based stress in the present study may be because both structured and free-form coloring induce a state of mindfulness, which refers to an awareness or attentional focus of being in the present moment. Indeed, drawing has been referred to as “creative mindfulness”, possibly because the individual is absorbed and engaged with what they are doing (Greenhalgh, 2016). Mindfulness therapy has been found to increase brain activation in areas responsible for increased adaptive responses to stressful situations (Cahn & Polich, 2006; Davidson et al., 2003), and decreased stress levels in therapists, medical students, and college students (see Davis and Hayes, 2011, for a review). It is possible that in the present study, both structured and free-form coloring allowed veterans with PTSD the opportunity to focus and creatively engage in the present moment, which resulted in a calm state.

Finally, the present study also found that free-form coloring, but not structured coloring, improved working memory in those with and without PTSD. Previous research has looked at related art activities, such as doodling, as a way to maintain attention to a task and improve memory (Andrade, 2009). However, this is the first study to explore the potential cognitive benefits of coloring or drawing. There are several possibilities for the reported working memory benefits. One explanation is that the lack of structure in the free-form coloring required additional cognitive resources to plan their drawing and take steps to accomplish what they wanted to achieve creatively within a given time-limit. Another possibility is that creating art requires abstract thinking and conceptual understanding that involves working memory (Lusebrink, 2004); also Alloway, Southard, Frankenstein, and Guess, 2017). Patients with PTSD typically show a decline in working memory (Honzel et al., 2014). Thus, practical and easy-to-execute interventions, such as drawing, may be a useful way to enhance working memory. Benefits of improved working memory include better planning skills, goal-directed behavior, and better multi-tasking skills (Alloway & Alloway, 2013).

There are some limitations to the present study. First, the classification of PTSD in participants was with the use of a short screener. While the Primary Care PTSD Screen was designed for use in primary care and other medical settings, it is not meant for diagnostic purposes. Positive responses on this screener should be followed up with further assessment and a structured, clinical interview. Thus, the use of “PTSD” for participants in the present study is not a diagnostic label, but a classification based on their self-reports on the screener. The second limitation is that although there were positive benefits of a short art-therapy, PTSD can be a long-term psychopathological condition. As with any new therapeutic activity, the long-term benefit should be addressed in future research, particularly in light of individual differences, such as race, gender, and trauma type.

Art therapy is a broad term that encompasses varying types of art media. This variability can pose some difficulty in assessing its efficacy in reducing acute stress (see Reynolds, Nabors, & Quinlan, 2000). Furthermore, art therapy is typically unstructured, in that the individual is encouraged to project their experiences on to their artistic productions. While this provides therapeutic value for the individual, it can be challenging to systematically evaluate potential progress. Tracks for future research include the use of a structured art activity, such as the approach in the present study, combined with a standardized measurement system of two-dimensional art called the Formal Elements Art Therapy Scale (FEATS; Gantt, 2001). The value of using this structured ratings scale is that it can be used to make differential diagnoses, measure therapeutic responses, and chart progress over time. In summary, the present findings suggest that simple art activities that can be done on an individual basis may yield both mental health and cognitive benefits, especially for veterans with PTSD.

Conflict of interest statement

Conflict of interest

There is no conflict of interest

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Intellectual property (patents, copyrights etc.)

There are no patents or copyrights

Supplementary materials

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